Change Detection and Impression Updating in Associative Memory for Face-Adjective Pairs

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1 Introduction

Aging, Proactive Interference, and Memory for Change

- Older adults exhibit more proactive interference in associative memory than young adults (e.g., Hasher & Zacks, 1988; Hay & Jacoby, 1999)
- Due in part to decreased memory for how associations changed across episodes (e.g., Wahlheim, 2014; Wahlheim & Zacks, 2019)

Impression Formation and Updating

- In social cognition, individuals are associated with attributions of moral character (e.g., Uleman & Kressel, 2013)
- Impressions may need to be updated in order to remain adaptive (e.g., Siegel et al., 2018)

Research Questions:

- Do age groups differ in memory for change re: personal traits?
- Does memory for change influence impression updating in young and older adults?

2 Method

Participants

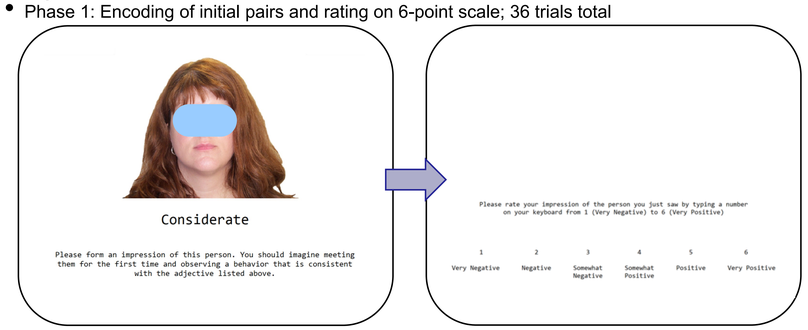
- 41 older adults (OAs; 65-91 years, M = 75.56 years; 12-24 years of education, M = 16.49 years) recruited from community around Elon University
- 36 younger adults (YAs; 18-21 years, M = 18.57 years, 13-16 years of education, M = 13.44 years) students recruited through Psychology Dept. participant pool

Stimuli

- 48 neutral faces (Minear & Park, 2004)
- 78 trait adjectives (half positive, half negative; Gruhn, 2016)

Design

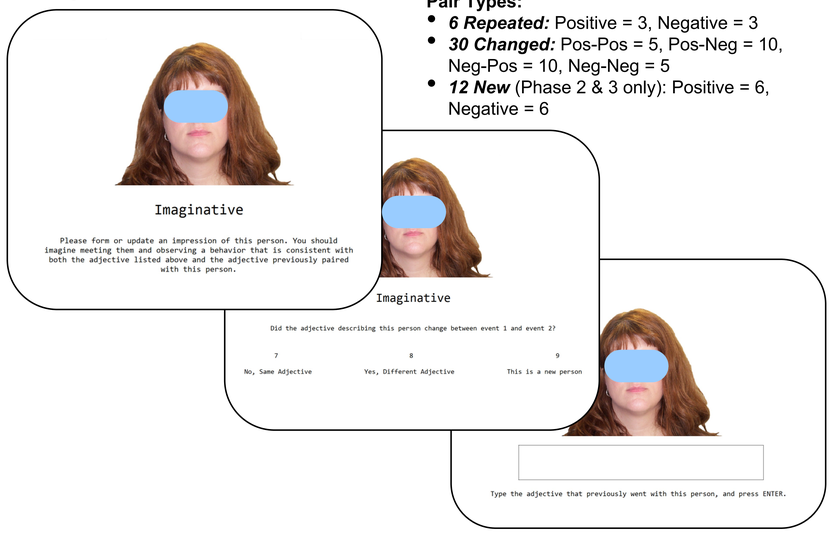
- Phase 1: Encoding of initial pairs and rating on 6-point scale; 36 trials total



- Phase 2: Encoding of same/different/new pairs, change detection, and recall of prior adjective, if change indicated; 48 trials total

Pair Types:

- ***6 Repeated:*** Positive = 3, Negative = 3
- ***30 Changed:*** Pos-Pos = 5, Pos-Neg = 10, Neg-Pos = 10, Neg-Neg = 5
- ***12 New*** (Phase 2 & 3 only): Positive = 6, Negative = 6



3 Method (cont.)

- Phase 3: Cued recall of Phase 2 adjective, memory for prior change, recall of Phase 1 adjective (if change indicated), and final impression rating; 48 trials total

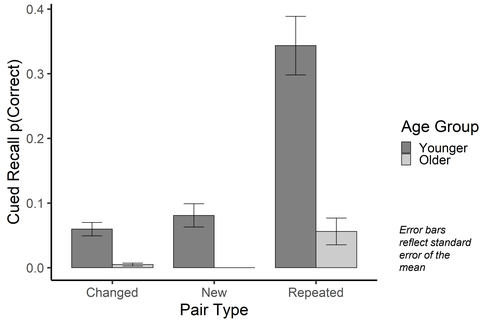
4 Results

Exclusions

- Data excluded from 8 OAs due to low MoCA scores; 3 YAs and 5 additional OAs due to technical difficulties or non-compliance

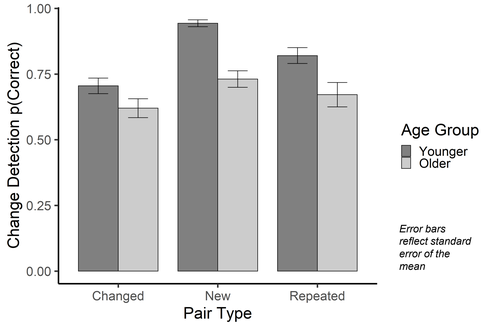
Memory for Adjectives at Test: Cued Recall Performance

- Recall (at Phase 3) of Phase 2 adjectives was very poor
- Only condition with a substantial proportion of correct responses was Repeated condition for YAs
- Recall of Phase 1 adjectives (in Phases 2 and 3; not shown) was also poor



Change Detection at Second Encoding

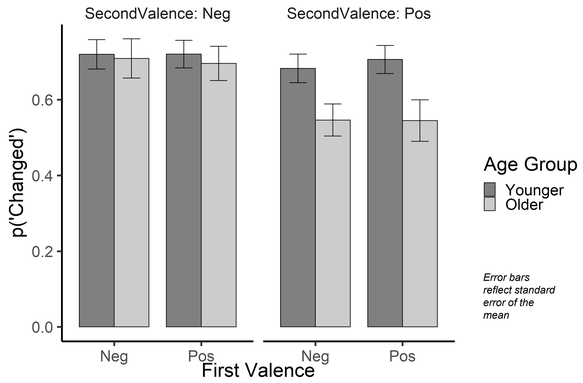
- At Phase 2, participants were reasonably good at identifying whether face-adjective pairs were repeated, changed, or new
- OAs had worse accuracy overall
- Classical ANOVA: effects of Age Group, $F(1,59) = 30.9$, $p < .001$; and Pair Type, $F(1.8, 106.9) = 15.6$, $p < .001$; no interaction, $F(1.8, 106.9) = 2.08$, $p = .14$ (Greenhouse-Geisser corrected)
- Bayesian ANOVA: Age + Pair Type (best model), $BF_{10} = 2.4e+8$



5 Results (cont.)

Change Detection at Second Encoding (cont.)

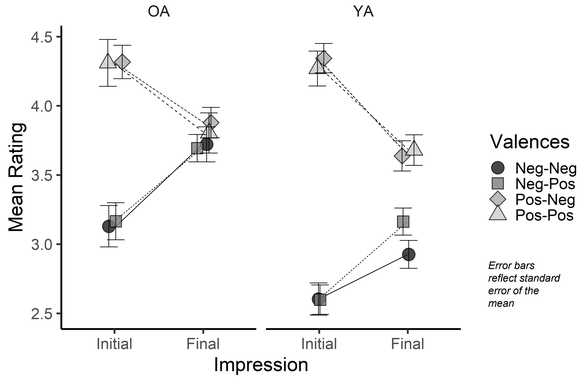
- Age difference in detection of changed adjectives depended on valence of the Phase 2 adjective
- OAs more likely to respond that adjective had changed when new adjective was negative
- Classical ANOVA: effects of Second Valence, $F(1,59) = 22.6$, $p < .001$; and Age Group x Second Valence, $F(1,59) = 11.7$, $p = .001$



- Bayesian ANOVA: Second Valence + Age + Second Valence*Age (best model), $BF_{10} = 841$

Initial and Final Impression Ratings

- Final ratings more neutral than initial ratings; may reflect forgetting of adjectives
- OAs' ratings more positive, particularly for faces paired with negative adjectives
- Classical ANOVA: effects of Age Group, $F(1,59) = 10.6$, $p = .002$; Valences, $F(1.9,112.9) = 77.7$, $p < .001$; Valences x Age Group, $F(1.9,112.9) = 6.36$, $p = .003$; Valences x Impression, $F(1.9, 110.8) = 50.7$, $p < .001$



- Bayesian ANOVA: Valences + Impression + Age + Valences*Age + Valences*Impression (best model), $BF_{10} = 2.5e+61$

6 Conclusions

- Associative memory effects on impression formation and updating are under-studied
- In the present experiment, both young and older adult participants had difficulty recalling specific adjectives paired with faces, but were able to detect when adjective changed
- OAs' change detection was as good as YAs when second adjective was negative
- OAs' change detection responses were likely influenced by positivity bias: when memory fails, OAs may assume prior adjective was positive
- OAs' average impression ratings quickly revert to mildly positive; YAs retain initial impressions longer, particularly negative
- Further research should examine OAs' impression updating under more favorable memory conditions

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